JUN SEOK KIM

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SUMMARY

- **MSc researcher** in uncertainty calibration for 3D perception, first-author paper targeting CVPR 2027.
- **Founder of TARS**, which runs monocular depth on edge cameras for 60+ paying clients.

RESEARCH INTERESTS

- Uncertainty calibration for learned perception
- 3D scene representation for robotics
- 3D perception under compute and communication constraints
- Real-time reconstruction of dynamic scenes

EDUCATION

The Chinese University of Hong Kong, Shenzhen Sep 2024 – Expected Mar 2027
MSc in Artificial Intelligence and Robotics | Capstone Project I: LocalAggTR, graded Distinction

Kangwon National University Mar 2017 – Feb 2023
BSc in Information Statistics | BSc in Engineering, Convergence Program of Data Science
Main Courses: Machine Learning, Theory of Probability, Multivariate Statistical Analysis

PUBLICATIONS & PREPRINTS

Range-aware uncertainty calibration in self-supervised monocular depth
Jun Seok Kim, Rui Huang
Manuscript in preparation, targeting CVPR 2027. Preprint expected November 2026. (project page)

RESEARCH EXPERIENCE

Computer Vision Laboratory, CUHK-Shenzhen, advised by Prof. Rui Huang Sep 2024 – Present

Range-aware uncertainty calibration in self-supervised monocular depth (project) Jan 2026 – Present

- Many robotic systems act on the magnitude of learned depth uncertainty, yet the uncertainty of self-supervised depth models is almost always evaluated by ranking, so that magnitude goes unchecked. On KITTI, MonoProb (WACV 2024) underestimates its error at every range, with error reaching 3.7× the level its predicted uncertainty implies.
- Showed that self-supervised retraining alone cannot close this gap, while a single label-free scale constant estimated from stereo residuals can. Retrained the model by self-distillation from MonoProb, training the uncertainty toward its disagreement with a second self-supervised depth model, then applied that constant.
- **Cut worst-range calibration error by 86%** versus the original model and by 67% versus rescaling fitted on LiDAR ground truth, using no ground-truth depth.

LocalAggTR: Differentiable Voxelization for Weakly Supervised 3D Occupancy Sep 2025 – Dec 2025
Capstone project (code)

- GaussTR's Gaussian-to-voxel step is non-differentiable and dominates inference time, so 3D occupancy losses cannot train the network end-to-end.
- Replaced it with a differentiable Local Aggregation operator that passes occupancy gradients.
- Made inference **7.2× faster with mIoU unchanged** on nuScenes. Adopted by the lab's 3D Gaussian Splatting project for semi-supervised fine-tuning.

Point Cloud Density and 3D Object Detection (code) Jun 2025 – Jul 2025

- Asked how a 3D object detector degrades when LiDAR density drops and whether upsampling recovers it.
- Built a resampling pipeline that thins LiDAR points by random or voxel-grid downsampling and restores them with two upsampling methods, then evaluated PointPillars on nuScenes-mini.
- Found that **how points are removed matters more than how many**. Voxel-grid thinning to one fifth of the points cost 2.1% of mAP, while random thinning to half cost about 4× more.

Geometric vs. Generative 3D Augmentation for Point Cloud Classification (code) Jan 2025 – May 2025

- Tested whether text-to-3D generative augmentation outperforms geometric augmentation.
- Built Point-E generation and geometric augmentation modules and ablated them with PointNet on ModelNet40.
- Geometric augmentation **raised accuracy by 6.5 points**, while text-to-3D generative augmentation did not help.

INDUSTRY EXPERIENCE

Founder Feb 2024 – Present
TARS, South Korea

- Built and operate a foot-traffic analytics service for restaurants, cafés and real-estate offices. Edge cameras use monocular depth estimation to separate storefront pedestrians from those across the street and measure hourly pass-by, stop and entry rates.
- Serve **60+ paying clients** with consulting reports based on this data. Self-funded and profitable since launch.

Backend & Infrastructure Engineer Jul 2022 – Aug 2022
Bluzen Corp., Singapore

- Built a MariaDB and Node.js backend for workforce records and networked Hyundai Motor server rooms.

Data Scientist Jun 2021 – Jun 2022
Connple Corp., South Korea

- Built and shipped a BERT-based restaurant recommender to a production mobile app, paired with SVD collaborative filtering on review ratings.
- Extracted food keywords from Instagram reviews with LDA topic modeling to power a food discovery service.
- Automated nationwide collection of 600,000 restaurant records through a scheduled Flask data pipeline.

SCHOLARSHIPS & HONORS

- Mid-Sized Company Employment Support Scholarship, Korea Student Aid Foundation 2021 – 2023
- Passion Award, 21st and 22nd Gangwon Province University Student Entrepreneurship Competition 2020, 2021

LEADERSHIP & TEACHING

- Teaching Assistant, Introduction to Python, Kangwon National University Mar 2020 – Jun 2020
- Team Lead, Student Startup Teams, Kangwon National University Jan 2020 – Dec 2021
- Squad Leader, 930th Artillery Battalion, Republic of Korea Army Apr 2018 – Nov 2019

SKILLS & LANGUAGES

- Programming: Python, C/C++, CUDA, SQL
- Frameworks: PyTorch, MMDetection3D, MMEngine, scikit-learn
- Tools: Git, Linux, Docker
- Languages: Korean (native), English (full professional), Mandarin Chinese (basic)